

ECE 486: Lab 4 Report

UNIVERSITY OF
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A Report Prepared For:

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ECE486

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Part 1 (20 marks)

Before the experiment, the camera was calibrated. We fixed the camera, then took more than ten pictures in different angle under the camera with the given black-white chessboard paper, and obtaining the camera matrix and distortion coefficients. And the camera-to-robot transformation was recalculated using the calibration procedure developed in Lab 3. To confirm successful camera calibration, we retested the error using the points in Lab 3. Since the initial error was relatively large, we recalibrated the camera and recalculated the transformation matrix several time, until we get following results.

Camera matrix:

$$C = \begin{bmatrix} 942.06711744 & 0 & 278.79611215 \\ 0 & 954.98658013 & 268.68056973 \\ 0 & 0 & 1 \end{bmatrix}$$

Distortion coefficient:

$$D = \begin{bmatrix} -0.03197389 \\ 0.61666864 \\ 0.00632412 \\ -0.00394646 \\ -2.95634379 \end{bmatrix}$$

Transformation matrix:

$$R = \begin{bmatrix} -0.97389296 & -0.09716424 & 0.2051624 \\ -0.13530187 & 0.97414712 & -0.18091656 \\ -0.18227975 & -0.20395222 & -0.96186152 \end{bmatrix}$$

Translation vector:

$$T = \begin{bmatrix} 0.18643354 \\ 0.16327346 \\ 0.43956406 \end{bmatrix}$$

Under these parameters, rerunning the validation code in Lab 3 shows that the error at all test points is below 8mm, this error is even lower than the error we got during our Lab 3. Multiple calibration attempts were performed prior to the final experiment. After each calibration, the verification procedure from Lab 3 was repeated to assess conversion accuracy. This report presents only the final calibration results, as these were used to label the experiments.

After calibration, we were given a sheet of paper with three ArUco marks. The camera automatically detects the center of the three ArUco markers as the target points, and we need to adjust the paper position to ensure that all three target points are within the robot's workspace. Due to the large size of the paper, our test run was rejected multiple times because the target points were detected outside the workspace. Since the default robot workspace was previously too small, under Dr. Cooper's instructions, we changed the size of the robot's workspace to ensure that the entire paper was within its range.

Restricted Lab 4 Workspace:

$$139.84 < x < 247.43$$

$$-56.57 < y < 88.66$$

Camera marker coordinate and calculated robot coordinate

Marker	Camera Coordinate	Robot Coordinate
1	[0.0848, 0.0393, 0.6276]	[145.3, 88.66, -206.12]
2	[0.0969, -0.1055, 0.6326]	[139.84, -56.57, -201.99]
3	[-0.0123, -0.0609, 0.642]	[247.43, -7.31, -200.39]

We first tested the robot without the pen attached to ensure safety. We set the height parameter to a value that would prevent it from hitting the table, and then let the robot to move to the center of the detected ArUco mark while maintaining a safe distance from the paper. This step aimed to verify that the calibrated parameters correctly mapped the detected mark positions to the robot's coordinate system.

Once we confirmed the robot could reach the correct marking positions, we installed the pen and manually measured the pen height, inputting the height limits to ensure the pen tip could move safely above the paper and allow approximately 5 mm of compression upon contact, meeting Lab 4 safety requirements.

Then, we instructed the robot to move above each mark, after reach the horizontal position, lower the pen tip 5mm vertically to contact the paper, mark one point, return to a safe height, and move to the next marker position, until all three points were marked. Throughout this process, we ensured the camera, robot, and paper remained stationary. Afterward, we manually measured the horizontal and vertical offsets (dx and dy) between the mark center and the pen tip using a ruler.

The positioning error for each mark was calculated in Euclidean distance $e = \sqrt{dx^2 + dy^2}$.

Although the lab manual suggested that multiple iterations of correction might be necessary, our group did not perform any additional positional corrections in Lab 4. Before the marking experiment, we repeated the camera calibration steps from Lab 3 until the verification error was below 8 mm. After the first marking experiment, we showed the marked paper to Dr. Cooper for review, who deemed the achieved accuracy sufficient for the experimental objectives. Therefore, we did not further correct the robot trajectory; this report only reports the final positioning error.

We summarized and analyzed three possible main sources of error.

1. Camera Calibration and Camera-to-Robot Transformation

Although we repeated camera calibration before the experiment, minor errors still may remained in the estimated camera intrinsic parameters and the camera-to-robot transformation matrix. These errors directly affected the calculated robot target coordinates, causing the robot's position to deviate slightly from the actual marker center.

2. ArUco Marker Detection

The robot target position was determined by the detected ArUco marker center. The detected marker center was affected by image resolution, lighting conditions, camera noise, etc. Even

minor errors in the detected marker position were transmitted through coordinate transformation and become positioning errors on the paper.

3. Paper Movement During Marking

During the marking process, the pen tip occasionally slightly pushed the paper when it touched the paper surface. Since the positions of all marker points are calculated based on initial camera images before the robot begins to move, any movement of the paper will change its position relative to the robot. As a result, the remaining target coordinates no longer perfectly match the actual marker positions, introducing additional positioning errors in subsequent marking. This effect mainly influenced the remaining target points after the paper had shifted.

We can improve the camera calibration process to further reduce errors. In the current calibration process, the camera's intrinsic parameters are estimated by capturing multiple images of the black-white chessboard and the chessboard remains roughly flat on the table. Although the chessboard is moved to different positions, the images are primarily collected from the same plane, with limited variations in vertical distance. Therefore, the calibration may not fully capture the camera's viewpoint variations throughout the workspace.

Furthermore, the camera-to-robot transformation is computed using 10 points in workspace. Although these points are sufficient to obtain a transformation matrix, the relatively small number of calibration points makes the results more sensitive to individual measurement errors.

To improve the calibration, the chessboard should not only be captured at different horizontal positions but also at different height relative to the camera. This will provide a broader viewpoint and allow for more accurate estimation of the camera's intrinsic parameters. Additionally, more corresponding camera and robot points should be collected when computing the rotation matrix and translation vector. The calibration points should not rely solely on ten points but should be distributed throughout the available workspace, including points near the boundaries and at different heights.

Part 2 (10 marks)

After the marking experiment, the horizontal (dx) and vertical (dy) offsets between the center of each ArUco mark and the corresponding pen marker were manually measured using a ruler. The resulting positioning errors are summarized in table.

Marker	dx (mm)	dy (mm)	e (mm)
1	4	6	7.21
2	4	4	5.66
3	5	0	5

Since no trajectory correction was performed after the initial marking experiment, no intermediate correction-error table is available.

The measured position error for 3 markers are 7.21 mm, 5.66 mm, and 5 mm, with an average error of 5.96 mm. Considering the entire vision-guided operation, the robot was able to reach all three target locations with good repeatability.

During the experiment, only the initial image was used to determine the marker positions. Taking new images while the robot is moving is not recommended, as the robotic arm or pen may partially obstruct the camera's view of the ArUco markers, leading to unstable or incorrect marker detection. Furthermore, repeated image captures may introduce minor biases due to image noise and changes in lighting.

However, using only the initial image also has potential errors. If the paper moves slightly during marking, or if the camera or robot's position changes after the first image is taken, the stored marker coordinates will no longer represent the true location. Since no additional images are taken, these changes cannot be detected, and the localization error will persist until the end of the experiment.